

Data exfiltration via Ransack query injection

Data exfiltration via Ransack query injection - Author not stated, GitHub Advisory Database.

- Published: 2023-01-09
- Original: <<https://github.com/openSUSE/travel-support-program/security/advisories/GHSA-2wwv-c6xh-cf68>>
- Preserved from: <https://github.com/openSUSE/travel-support-program/security/advisories/GHSA-2wwv-c6xh-cf68> (github-api) on 2026-10-02
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Content

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Data exfiltration via Ransack query injection

- Advisory: GHSA-2wwv-c6xh-cf68
- CVE: CVE-2022-46163
- Severity: high
- Published: 2023-01-09
- Updated: 2023-01-09

Affected

- `travel-support-program` (deployment): <d22916275c51500b4004933ff1b0a69bc807b2b7

Description

Impact

Sensitive user data (Bank Account Details, Password Hash) can be extracted via Ransack query injection. Every deployment of `travel-support-program` below the patched version is affected.

The `travel-support-program` uses the Ransack library to implement search functionality. In its default configuration, Ransack will allow for query conditions based on properties of

associated database objects [1]. The `*_start`, `*_end` or `*_cont` search matchers [2] can then be abused to exfiltrate sensitive string values of associated database objects via character-by-character brute-force (A match is indicated by the returned JSON not being empty). A single bank account number can be extracted with <200 requests, a password hash can be extracted with ~1200 requests, all within a few minutes.

Patches

The problem has been patched in commit `d22916275c51500b4004933ff1b0a69bc807b2b7`

Workarounds

In order to work around this issue, you can also cherry pick that patch, however it will not work without the Rails 5.0 migration that was done in #150, which in turn had quite a few pull requests it depended on.

References

[1] <https://activerecord-hackery.github.io/ransack/going-further/associations/> [2]
<https://activerecord-hackery.github.io/ransack/getting-started/search-matches/>

Credit

The vulnerability was reported to us by Lukas Euler from Positive Security. Thank you!